Assignment 3 — Morphology, Edges, Frequency & Shape (Applied)

Rule of thumb for students:

If you can't show it visually, you don't understand it yet.

Tools allowed: Python, NumPy, OpenCV, Matplotlib
Submission: Single Jupyter Notebook

Task 1 — Binary Image + Structuring Element

What to do

1. **Load any grayscale image**
2. **Convert it to a binary image using thresholding**
3. **Create three structuring elements:**
   - **3×3 square**
   - **3×3 cross**
   - **5×5 square**

Output

Display:

- **Binary image**
- **Result of dilation using each SE (side by side)**

Constraint

Do not reuse the same SE shape twice.

Observation (1–2 lines in comments):
How does changing SE shape affect object growth?

Task 2 — Noise Removal Without Touching the Object

What to do

1. **Add salt noise to the binary image**
2. **Remove noise using only morphological operations**
3. **You may use:**
   - **Erosion**
   - **Dilation**
   - **Opening**
   - **Closing**

Output

Display:

- **Noisy image**
- **Cleaned image**

Constraint

❌ **No blurring**
❌ **No median filter**

Observation:
Which operation was most effective?

Task 3 — Fill Gaps in Broken Objects

What to do

1. **Take a binary image where objects have small holes or breaks**
2. **Use morphology to:**
    - **Fill holes**
    - **Preserve object size as much as possible**

Output

Display:

- **Before**
- **After**

Observation:
What happens if SE size is too large?

Task 4 — Sobel vs Canny (Edge Reality Check)

What to do

1. **Convert a natural image to grayscale**
2. **Apply:**
    - **Sobel edge detection**
    - **Canny edge detection**

Output

Display:

- **Original image**
- **Sobel result**
- **Canny result**

Constraint

Use at least two different Canny threshold pairs

Observation:
Which edges disappear first when thresholds change?

Task 5 — Line Detection from Scratch (Pipeline Thinking)

What to do

1. **Use the Canny output**
2. **Apply Hough Line Transform**
3. **Draw detected lines on the original image**

Output

Display:

- **Canny edges**
- **Final image with detected lines**

Constraint

Do not tune blindly — change one parameter at a time

Observation:
What happens when the threshold is too low?

Task 6 — Fourier Transform: See the Invisible

What to do

1. **Compute the 2D Fourier Transform of a grayscale image**
2. **Display the magnitude spectrum**

3. **Apply:**
    - **Low-pass filtering**
    - **High-pass filtering**
4. **Reconstruct images using inverse FFT**

Output

Display:

- **Original image**
- **Spectrum**
- **Low-pass result**
- **High-pass result**

Observation:
Which operation affects edges more?

Task 7 — Contours as Object Boundaries

What to do

1. **Threshold an image**
2. **Detect contours**
3. **Identify and draw:**
    - **All contours**
    - **Largest contour only**

Output

Display:

- **Contours overlaid on original image**

Constraint

Do not use edge detection before contour extraction.

Observation:
What happens if thresholding is poor?

Final Challenge — Combine Everything

What to do

Using any image of your choice:

1. **Use morphology to clean it**
2. **Detect edges**
3. **Extract contours**
4. **Highlight only meaningful object boundaries**

Output

A single final image showing clean object outlines.

Submission Rules (Strict)

- **One `.ipynb`**
- **Each task in a separate section**
- **Outputs must be visible**
- **Observations must be short, practical, and honest**